# Title

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**Abstract**

Maximum 250 words.
**Keywords:** keyword1, keyword2, keyword3.

## 1 Introduction

Context, motivation, and objectives of the work. The manuscript must be written in English or Portuguese and have between 10 and 20 pages.

The written work must be delivered until 7th of December 2026 and it must be orally presented in the seminar classes of the 1st or 3rd of December 2026 (following the schedule given opportunely). The presentation must be clear and objective, with a maximum of 10 minutes for the presentation. The written work must be delivered in PDF format, together with the source codes.

The grade of the discipline will be based on the written work (70%) and the oral presentation (30%). Aspects such as clarity, objectivity, and adequacy of the work to the course content [**?**] will be considered in the evaluation. The work must be developed individually and the use of AI tools must be properly cited.

### 1.1 Application setup

Describe the application setup. **The application setup must be presented to the professor in a clear and objective way until 15th of October 2026**.

## 2 Methods and techniques

Describe the methods and techniques used in the work. The work must be an application of the concepts learned in the course, and must involve at the minimum one of the following topics: linear systems, optimization, eigenvalues and eigenvectors, advanced integration.

## 3 Results

Specific tests and results.

## 4 Conclusions

Present the conclusions of the work.

## References